

# **Revised Generation Interconnection System Impact Study Report**

## **For Queue Project Q13-0265**

North Little Rock Levy 138 kV  
September 18, 2015

### ***1. Introduction***

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0265. The Interconnection Customer proposes a 170.0 MW wind facility interconnecting at the North Little Rock Levy 138 kV substation in the Tennessee area. The request was received on August 26, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 144 higher-queued active requests and their assigned network upgrades.

### ***2. Summary of Results***

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,461,000.

### ***3. Network Impacts***

#### ***3.1 Generator Deliverability and Single Contingency Analysis***

For the loss of Elberfeld S to Boulevard 138.0 kV, the N Little Rock Levy to East Sanger 138.0 kV line loads to 110.7% of its 216 MVA rating. Project DFAX: 69.0%.

The East Sanger to Truman Tap 138.0 kV is overloaded to 108.2% of its 210 MVA emergency rating for the loss of the Elberfeld South to Boulevard 138.0 kV line. The project contributes 69.0% DFAX (117.3 MW) to this overload.

For the loss of Elberfeld S to Boulevard 138.0 kV, the Truman Tap to Mocapueblo 7101-704 138.0 kV line loads to 106.3% of its 208 MVA rating. Project DFAX: 69.0%.

Boulevard Substation to Elberfeld S 138.0 kV: 105.4% loading (239 MVA) following the outage of the East Sanger to North Little Rock Levy 138.0 kV; DFAX = 61.4%.

For the loss of East Sanger to N Little Rock Evy 138.0 kV line, the Boulevard to W Jax 138.0 kV loads to 105.0% of its 231 MVA rating. Project DFAX: 61.4%.

West Jax to Rockwood 138.0 kV: 102.6% loading (226 MVA) following the outage of the East Sangr to North Little Rock Levy 138.0 kV; DFAX = 61.4%.

For the loss of North Little Rock Levy to E Sanger Substation 138.0 kV Line, the Stonyhill to Rockwood 138.0 kV loads to 100.3% of its 211 MVA rating. Project DFAX: 61.4%.

#### ***3.2 Multiple Facility Contingency Analysis***

No additional overloads were identified for multiple facility contingencies.

#### ***3.3 Short Circuit and Stability***

Short circuit and stability analyses identified no additional network upgrades.

### ***4. Required Network Upgrades and Cost Estimates***

| Facility                                        | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| N Little Rock Levy to East Sanger 138.0 kV line | Rebuild line with higher-capacity conductor | \$9,191,000   | 100% (\$9,191,000)    |
| E Sanger to Trumn Tap 138.0 kV Line             | Reconductor line                            | \$3,869,000   | 100% (\$3,869,000)    |
| Truman Tap to Mocapueblo 7101-704 138.0 kV line | Reconductor line                            | \$3,982,000   | 100% (\$3,982,000)    |
| Boulevard Substation to Elberfeld S 138.0 kV    | Reconductor line                            | \$1,844,000   | 100% (\$1,844,000)    |
| W Jax to Boulevard 138.0 kV                     | Reconductor line                            | \$1,103,000   | 100% (\$1,103,000)    |
| West Jax to Rockwood 138.0 kV                   | Reconductor line                            | \$4,359,000   | 100% (\$4,359,000)    |
| Stonyhill to Rockwood 138.0 kV                  | Reconductor line                            | \$2,113,000   | 100% (\$2,113,000)    |
|                                                 | Total allocated to project                  |               | \$26,461,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.